

Adoptive $\gamma\delta$ T cell therapy controls cytomegalovirus infection in preclinical models relevant to transplantation.

Corresponding Author: Professor Lionel Couzi

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

In the manuscript by Marsères et al, a new cell therapy is proposed to combat the important issue of post-transplantation refractory CMV infection in donor organ recipients. Especially CMV-naïve recipients receiving a transplant from CMV-seropositive donors are at high risk of developing refractory CMV infection, ultimately leading to diminished transplant survival outcomes. Adoptive transfer of CMV-specific $\alpha\beta$ T cells has previously demonstrated benefits for graft survival, but this cell population is challenging to obtain from CMV-naïve recipients and would require HLA-matching in an allogeneic setting. To circumvent these challenges, adoptive transfer of HLA-independent $\gamma\delta$ T cells is proposed, of which specifically the V δ 2neg $\gamma\delta$ T cell population has demonstrated anti-CMV immune response, also called "Delta One T" or DOT cells. First, the authors demonstrate the suitability of a scalable production method for clinical-grade DOT cell expansion, capable of expanding DOT cells both from CMV seropositive and seronegative healthy donors with comparable effector characteristics. The authors demonstrate that CMV-mediated IFN- γ induction is dependent on LFA-1 and independent of the $\gamma\delta$ TCR presence on the DOT cells. Subsequently, the authors use a murine CMV infection model to recapitulate the human situation in an in vivo setting. Finally, the authors demonstrate DOT cell expansion is possible from CMV-naïve receivers of CMV-seropositive donor kidneys under immunosuppressive conditions, while maintaining CMV-induced IFN- γ secretion in these cells. Altogether, the authors suggest a robust proof of feasibility for DOT cell therapy in kidney transplant recipients is demonstrated with the evidence presented in the manuscript.

We have thoroughly reviewed the presented evidence, and although in general the level of evidence provided support the authors' conclusions, we do have some concerns that need to be addressed before we can provide a positive opinion on accepting the manuscript in a high-level journal such as Nature Communications. Our concerns are illustrated below.

General considerations and points for improvement

1. Throughout the manuscript, the authors present experiments using different numbers of donors per experiment, and donors with different CMV-serological status. It is often unclear from the figure whether the same donors are used as in the figure before, and which criteria are used to select a specific donor (e.g. figure 3 B: 4 donors, 3C: 2 donors – are the 2 in C also represented in B, and if so, why are these 2 selected over the other 2 used in B?). Although the authors demonstrate comparability of DOT cell characteristics obtained from CMV naïve vs. CMV seropositive healthy donors in the first characterization experiments, in later experiments other characteristics are interrogated in more detail, in which differences between the serological status can become more apparent. By not disclosing the origin of the cells used in experiments after figure 1, these (likely minor) differences may be overlooked. Altogether, in our opinion the interpretability of the presented data would improve by disclosing this missing information.
2. In the materials and methods it is mentioned that, unless stated otherwise, IL-18 is used in the co-culture experiments. This creates confusion when going back and forth between main figures and supplementals, with the need to constantly re-read the legends searching for a mentions of IL-18 absence before being able to compare figures. In our opinion, adding a simple 'IL-18+' or 'IL18-' label in the figures would improve readability and interpretability.
3. Many abbreviations and acronyms are used, which are not consistently explained (sometimes in the main text, sometimes only in the figure legend). We urge the authors to critically go through the main text and explain abbreviations/acronyms appropriately. Highlighting (bold letters) and using a glossary would increase readability, if this is allowed in the Nature Comms format.

4. The figure panels (especially the flow cytometry gates panels) are too small to read comfortably from an A4 printed paper format. We strongly advise to consider using larger panels.

5. The authors address the risks of off-target reactivity in the discussion. What is the expected side effect profile in humans, this should be considered. More data are needed, not only mock infected cell lines (which also do not induce 'zero' response) and a reference to an ongoing clinical study in leukemia patients. The SOT patient with an ongoing infection (activated stage, possibly other viruses) and the TCR independent mechanism as shown in this manuscript should be taken into account and when possible other targets should be included in the experiments.

Specific considerations

Methods section: no mentioning of off target scores of the used sgRNAs for the KO model. Was this check performed? How else have off-target editing effects been excluded? Please explain.

Line 375: Please elaborate on how these donors are selected and is this enough to make the conclusion in line 390-391?

Figure 1E: include CMV status (donor 1-2 = CMV neg, donor 3-4 = CMV pos)

Figure 1F: In the text the terms central memory or effector cells are used, in the legend within the figure this is translated to CD27 and CD45RA status. Provide a label in the figure for the central memory or effector population as well for clarity. Also, SD are shown, but no statistical analyses are provided.

Figure 1F is associated with figure 5E (where a similar analysis is performed in the murine CMV in vivo model). We suggest to model the 2 figures in a similar way so they can be compared side by side and similarities in orthologous cell populations are immediately apparent.

Line 409-410: more elaborate explanation of why authors chose to focus on IFN- γ is needed. They refer to a paper, but a short explanation of why this is the most interesting factor (compared to the other targets in the CMV-induced secretome) would be appreciated.

Figure S3A: contains endothelial cells, effect of CMV infection is minimal. Please comment on why this is (and also why in S3D they don't seem to react to CMV induction without IL-18, but with IL-18, suddenly IFN- γ is increased).

Figure S3B: FMO of infected cells is missing. Can the addition of these cells lead to an increase in the CD107a fluorescence channel? These data should be added.

Line 412: IL-18 is used in most co-cultures (please add in the figure panel e.g. in S3D when it is used and when not), but how does this relate to an expected effect in vivo in a human? As it seems IL-18 is needed to increase functionality of the gd T-cells. Would you expect to see the same effects as observed in vitro, when in an in vivo situation you cannot co-stimulate with IL-18? Please comment on that.

Figure S3C: no statistical analyses. Different secretome factors are shown (with various efficiency of induction by CMV infection). IFN- γ was selected as the most important one. Was this based on statistical significance?

Figure 2D: please explain the use of HSV and VZV in more detail. Why are they used as controls? Are there controls used to show successful infection, what is their infection rate? Infection and expression of specific antigen (IE-1) is shown for CMV-infected cells, but not for HSV and VZV-infected cells. Is this the correct infection model these viruses? A bit of background is needed. And also are these still considered the right controls considering that later in the paper is shown that the cells do not require TCR engagement.

Figure 2 figure legend 'Mock-infected fibroblasts were used to set the GFP-positive gate': please show the gating used in the supplements.

Line 439-441: 'addition of a V δ 1 TCR blocking....' Could this be because the effect is mainly through V δ 3 then? Please comment.

Line 443: 'two distinct donors' – why are these donors selected and what makes them 'distinct' (in what respect? CMV serology?). What is their CMV Serological status?

Line 446: 'edition' is 'editing' meant here?

Line 450: explain briefly what the expected effect is of PMA ionomycin (used as a general inductor, this may be common knowledge in the field but requires a short explanation for the non-hematological experts reading this paper).

Line 452: remanent = remnant or better even 'remaining' ?

Figure 3A: Here fold increase is depicted while in the rest of the figure pg/ml IFN γ production is depicted. It makes it easier to read these figures when the axes are comparable.

Figure 3B: lacks a positive control for the blocking effect of $\alpha V\delta 1$

Figure 3D (and also in other figure panels where similar antibodies are used): it is unclear whether antibodies used are inhibitory or inducing. Please improve the colour scheme in a way to clearly show the expected effect, so it is clearly visible in the figure.

Figure 3G: 'representative' donor – only 1 donor is shown so this is an N=1 figure, while donor variability can be important here.

Line 472: it is not entirely convincing from the presented data that the main response is through LFA-1. We do not agree with the word 'mainly', difference between ICAM and LFA-1 is not convincing.

Line 475: edition = editing?

Figure 4D+E(bottom): because the KO of donor 2 is less efficient, there is a large spread in IFN induction in donor 2 and no difference between mock and CMV cocultured cells, so we are effectively looking at data from only 1 donor that is conclusive. Based on an N=1 we cannot draw these strong conclusions. Additional data is needed here.

Line 481 - 482: Not enough data here to conclude it is a key-mediator (see comment above).

Figure 5E: see remark for Fig 1E. Mirror these 2 so you can compare human and mouse.

Figure 5F: conclusion on LFA-1 cannot be drawn. For human, MdFI is measured in figure 1H to show LFA-1 is induced (in heat map this is comparable for day 0 and day 21. In 5F there is no remarkable difference in LFA-1 between day 0 and day 7. We cannot assume without the MdFI that the effect is the same as in human, so it cannot be said that (line 507) LFA-1 expression is upregulated.

Figure 5G: 2 doses of cells given how does this compare to the dosing regimen in human? Also: unclear label, in the top it says C57BL/6J +/- MCMV, is that the serological status? Because below it says 'MCMV i.p.' which is infection time point. The top reads as if the mice are administered the cell therapy and simultaneously receive the MCMV virus or not. Please change to less confusing label.

Figure 5H: the lethality differences are not very convincing. Plus there seems to be a difference between MCMVpos and neg (this is ns, but the numbers per group are different), any comments on this?

Figure 5I: this is a confusing experiment. It tells us that in mice that are still alive, the viral load in the organs is lower than in mice that were lethally infected. The mock treated mice (with RPMI) all had a high titer in the organs and were lethally infected. But of the treated mice, some still end up in the lethally infected groups with high viral load. Ergo, there seems to be a trend but you cannot draw a causative conclusion that the lower viral load is because of the cell therapy.

Figure 5J: Please comment on what the 'effector memory population' would be. In the text it is said the transferred T cells had a 'TEM phenotype' (line 521-522) but TEM is characterized as CD62L-, CD44- in the figure legend. The majority of cells is CD62L-, CD44+.

Line 524-525: this is a strong conclusion which is not entirely supported by the presented data mainly in figure 5H and I. In addition, it should be discussed whether this is the right mouse model to make these conclusions.

Title of last results paragraph (p24): we do not agree that the DOT cells 'meet cell therapy requirements'. This is a generalization. While the DOT cells may meet some requirements, the data presented does not unequivocally show it meets all the cell therapy requirements.

Line 544: please explain why it is necessary to add IL-15 to these cultures. Is IL-15 needed to make the cells less vulnerable for the immune suppressive drugs? And how does this relate to the in vivo situation?

Figure 6F and H: please comment on the very low proliferation index observed (high contrast with the DOT expansion protocol from healthy donors) and the relatively low amounts of live cells? H also doesn't have statistics.

Figure 6I: what is the dotted line representing? Mock+DOT as a baseline? Please explain?

Figure legend for 6 on p. 44. Inconsistent representation of 'donors' vs. 'experiments'. Sometimes donors are mentioned, sometimes it's 'representative experiments' but are these then from same donor? Please make this consistent and clear how many donors (patients or healthy) are used per experiment.

Reviewer #2

(Remarks to the Author)

The authors present interesting data regarding an adoptive gamma-delta T cell based approach using only 50 ml of patient

blood to tackle CMV disease which are of significance to the field, however to increase rigor, the amount of donors should be increased to show be more convincing.

I would recommend the following adjustments:

- l. 370: The numbers/percentages of TEMRA should be mentioned.
- At least 6 donors should be shown for each experiment, preferentially 3 females and 3 males, technical replicates are less relevant, they can be used to calculate a medium/median for each donor
- The appropriateness of the mouse experiment should be further discussed regarding differences of humans and mice gamma-delta T cells.
- l. 501: All data should be shown either in the supplement or main figures, not showing data is not acceptable.
- l. 528: Is there supporting evidence for this claim, especially having in mind the TRACE study?
- l.532: Actually, there are many reports showing successful expansion of autologous T cell products from kidney transplanted patients, please show evidence or adjust the statement (e.g. 10.3389/fimmu.2019.01148).
- l. 542: What was the rationale to exclude glucocorticoids from the study, although they are standard immunosuppressants used after SOT? Please add data on the effect of glucocorticoids on the cell product.
- Please add data on the killing capacity of patient derived T cell products regarding killing of the CMV+ fibroblasts.

Reviewer #3

(Remarks to the Author)

The paper by Marsères et al. asks whether the long-established capacity of human non-V δ 2+ T cells to respond to CMV-infected cells (and, at the same time, tumor cells) can be exploited as a cell therapy using the DOT cell technique to propagate and enrich for V δ 1+ T cells. Overall, the results seem encouraging and provide a basis for considering a clinical trial to limit CMV infection or reactivation in immunosuppressed transplant recipients. This is a major unmet need although analogous clinical approaches using $\alpha\beta$ T cells are further advanced. For the most part, the experiments are logical and seem well-executed, but there are areas where the authors' interpretations need to be placed very clearly in the context of DOT cells since they may not apply to human $\gamma\delta$ T cells in vivo. Indeed, this is an immunotherapy paper rather than a paper about human immunology, which is fine, so long as the two are not confused.

Main points:

Figure 1: this shows that DOT cells can be expanded equally well from CMV-sero(+) donor and CMV-sero(-) donors, thereby showing that an initial status of terminally differentiated effector non-V 2+ cells in CMV-sero(+) donors does not prevent the cells expanding and adopting a phenotype more like activated central memory cells. However, a major advantage of $\gamma\delta$ T cells for immunotherapy is their lack of MHC restriction and potential to be used in the allogeneic setting, in which case, a set of CMV-sero(-) donors would presumably be the main source. Hence, why is it important to know that cells from CMV-sero(+) donors can be expanded? The authors might argue that this is because autologous cells may be required for solid-organ recipients, but for that setting, which is a major unmet need, the authors need to be much more clear about the advantages over autologous $\alpha\beta$ T cells.

Figure 2: this shows that DOT cells from sero(+) or sero(-) donors limit CMV dissemination in vitro. However, the data do not seem as simple as the text describes. First, there is rather marginal activation of cytolysis by CMV-infected fibroblasts compared to medium alone, whereas the strong activation of cytolysis against infected macrophages is barely greater than that against mock-infected macrophages. For putative DOT-immunotherapy the fibroblast data might question efficacy, whereas the macrophage data might question safety. The IFN γ data seem more convincing.

The specificity for CMV versus other viruses is striking, but this needs to be considered in the context of Fig 3.

Fig 3: this shows that the TCR is not required for targeting CMV-infected cells. Note, the abstract states: "DOT cells efficiently controlled CMV dissemination in vitro. This antiviral effect was independent of the $\gamma\delta$ TCR". This seems very surprising given the wealth of evidence from this team that the TCRs of human non-V δ 2 T cells recognise self-antigens upregulated on CMV-infected cells. Furthermore, it was reasonable to assume that such TCR reactivities would account for CMV-specificity (above). So, do the data hold up? The use of a blocking antibody is problematic because without a clear control to prove otherwise, it is common for anti-TCR antibodies, even those that can block, to cross-link TCRs, thereby clouding the interpretation. Therefore, the logical experiment is to delete the TCR, which is accomplished. Here the problem may be that (according to the Methods) the TCR deletion was validated over a 4-day period after which cells were rested overnight, and then used in the experiments. Prior to this, the cells had constant TCR stimulation for over two weeks and probably therefore carry a legacy of this exposure in terms of downstream signalling effectors and epigenetic imprinting. This could explain why the evident reduction in IFN response (Fig 3E) is not complete. The hypothesis that TCR signalling is required could explain why there was no impact of blocking any NKRs (Fig 3G) and seems to be supported by the data in Fig 4 (below). Also, the argument that the high background of IFN γ in DOT cells responding to mock-transfected cells probably relates to a legacy of continued TCR signalling does not seem consistent with there being no background IFN γ in cells stimulated with isotype antibodies as a control for anti-CD3 (compare Fig 3D and 3E).

Fig 4: this claims to show that LFA1 is important for the response to CMV-infected cells. This would be consistent with the TCR engaging CMV-infected targets and activating LFA1 via inside-out signalling, thereby enhancing intimate effector-target cell contacts. One can hypothesise that in TCRk.o. cells, residual signalling events following exposure to chronic TCR signalling are sufficient to activate LFA1. Indeed, if the TCR was not involved, and none of the NKRs examined in Fig 3G is involved, it is very challenging to understand how activating LFA1 alone could induce a response from T cells, since it is at best a co-stimulator and primarily an adhesion molecule. It is perhaps important to note that if the TCR were not required, the authors would be obliged to explain why transfer of $\gamma\delta$ T cells would be preferable to transferring NK cells. In fact, the response of donor 2 LFA1k.o. cells (Fig 4E) is interpreted as negative, but it looks equally as strong as the responses of donors 1 and 2 TCRk.o. cells (Fig 3E) which were scored as positive. In sum, it seems highly likely to this reviewer that human V δ 1 cells require the TCR to engage CMV-infected cells and that they then activate LFA1. In DOT cells, LFA1 is already activated by chronic TCR signalling and can collaborate with residual TCR signals in TCR k.o. cells. This is an unusual situation that should not be confused with human immunology.

Fig 5: this shows that murine $\gamma\delta$ T cells produced via a DOT-type culture can be transferred to recipient lymphocyte, ILC, and NK cell deficient mice, wherein they become activated and distribute into multiple organs to limit CMV in vivo. This is a proof-of-principle; one suspects that NK cells might have achieved the same outcome.

Fig 6: this shows that the DOT protocol can be applied to kidney transplant recipients and that the cells may be functional in the presence of common immunosuppressants that would be the milieu for an adoptive therapy. This is encouraging although no evidence is given for how the cells remain functional in the immunosuppressants.

Reviewer #4

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Reviewer Response to Rebuttal Letter (Reviewer #1 + co-reviewer)

We would like to begin by congratulating the authors on the work they have put into revising their manuscript. The additional experiments, clarifications to the figures, and improvements to the text have significantly enhanced the quality of the work. We especially appreciate the consistency in which CMV status of the donor cells is now shown in the figures, providing transparency in the functional equivalence of the DOT cells expanded from different donors. These thoughtful revisions demonstrate a strong commitment to scientific rigor and communication, and we commend the authors for their efforts.

That said, we would like to kindly request the authors to address a few remaining points:

1. Inclusion of Reviewer-Specific Figures

Several figures created specifically in response to reviewer comments (the CRISPOR output data with sgRNA specificity scores, and specifically R1 and R3) provide valuable insights and clarity. We believe these should be included in the supplemental data to ensure transparency and to benefit future readers. Figure R1 addresses baseline gating data often requested in the field for assay validation purposes. Furthermore, figure R3 shows important data serving as a positive control for main figure 3B, without which the main figure is not adequately interpretable.

2. Clarification on Specific Findings

We would appreciate further commentary on the following points:

- Comment #14 and #39: While the authors have touched on the presence of IL-18 (in CMV-infected human endothelial cells) and IL-15 (in bone marrow of CMV-infected individuals), a more detailed contextualization would strengthen the author's views on what this means regarding the survival and activity of DOT cells in human patients. For instance, which concentration of these cytokines is required for the activity/support of these cells in vitro, and how does that correlate with expected levels in relevant tissues in a CMV-infected SOTR patient receiving the DOT therapy?
- Figure R3 and main figure 3B: We do not fully agree with the author's interpretation of these results. Although from later experiments using functionally knocked-out $\gamma\delta$ TCR cells, it can be concluded that DOT cells can recognize CMV-infection in the absence of $\gamma\delta$ TCR, the data presented in R3 and 3B are not contributing to this

finding. Specifically, in R3, the authors show the inhibitory effect of the V δ 1 antibody, which is reducing IFN γ induction by approximately 50% (and sometimes less). We appreciate the significant reduction compared to iso control, but the blocking effect seems incomplete, and residual δ 1 signaling effects cannot be excluded. Combined with the data presented in figure 3B, where we see variability of DOT cell response to CMV infection in combination with V δ 1 blocking antibody on the individual donor level, with most donors showing a decrease in IFN γ secretion after V δ 1 blocking, and 2 donors showing an opposite effect. Grouped together, these changes are not deemed statistically significant different ($P=0,051$), however in the context of incomplete V δ 1 blocking and the large variability observed, the validity of this conclusion can be debated (i.e. the lack of difference may be because residual δ 1 signaling is sufficient to maintain a certain level of IFN γ secretion)

- Figure 3H: This result raises an interesting observation that merits a brief discussion, even if speculative – now that the CMV status of the donor cells from which the DOTs are expanded is made visible, it seems that a trend exists of higher IFN γ induction in CMV + IL-18 cocultured DOTs from CMV negative donors (ie: gray dots are predominantly in the upper values, while the red dots dominate the lower values). Do the authors wish to comment on this interesting trend?

- Figure 3G: We appreciate the added clarity of activating and blocking antibodies to entire figure 3. Specifically in figure 3G, the activating label underneath ICAM-1 is missing. Please add the missing label for additional clarity.

- Lastly, we encountered these 2 minor points:

1. the HSCTR acronym is explained in line 49, but used earlier without an explanation in line 31 and 33.

2. We noticed that in the abstract and other places in the main text, 'amplified' is changed to 'expanded', which fits better in the context of proliferating cells. The authors should consider making this same change to line 65 and 117, where 'amplified' is still used.

We consider these points to be addressable, and we look forward to seeing them incorporated in the final version. Once again, we thank the authors for their work and responsiveness throughout the review process.

Reviewer #2

(Remarks to the Author)

All comments have been addressed in an appropriate manner.

Editorial Note

Due to the unavailability of reviewer 3 in this round, reviewer 2 also provided comment in their place on the response of the authors to the concerns of reviewer 2 in the previous round:

After thoroughly going through reviewer 3's concerns and the responses provided by the authors, I would conclude that the manuscript has been substantially improved and the concerns were appropriately addressed.

Reviewer #4

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

We confirm that the authors have addressed all the points raised in our previous review.

Reviewer #4

(Remarks to the Author)

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

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Point-by-point response

Reviewer #1 (Remarks to the Author):

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We have thoroughly reviewed the presented evidence, and although in general the level of evidence provided support the authors’ conclusions, we do have some concerns that need to be addressed before we can provide a positive opinion on accepting the manuscript in a high-level journal such as Nature Communications. Our concerns are illustrated below.

[Reply:](#) First, we would like to thank the reviewer for summarizing the main findings of our study, and for their careful reviewing of the manuscript. All the comments were thoroughly considered, and in most cases, complementary experiments were conducted in response. These efforts have substantially enhanced the quality of our results.

General considerations and points for improvement:

1. Throughout the manuscript, the authors present experiments using different numbers of donors per experiment, and donors with different CMV-serological status. It is often unclear from the figure whether the same donors are used as in the figure before, and

which criteria are used to select a specific donor (e.g. figure 3 B: 4 donors, 3C: 2 donors – are the 2 in C also represented in B, and if so, why are these 2 selected over the other 2 used in B?). Although the authors demonstrate comparability of DOT cell characteristics obtained from CMV naïve vs. CMV seropositive healthy donors in the first characterization experiments, in later experiments other characteristics are interrogated in more detail, in which differences between the serological status can become more apparent. By not disclosing the origin of the cells used in experiments after figure 1, these (likely minor) differences may be overlooked. Altogether, in our opinion the interpretability of the presented data would improve by disclosing this missing information.

Reply: Thank you for this fair suggestion. Because the number of DOT cells that can be expanded from one donor is limited, we generally used different donors for different assays. However, in some cases, it was possible to use the same donor in different assays. Therefore, in agreement with the reviewer and to clarify this point, we have now identified in the revised manuscript each donor with a specific number, which is indicated either directly in the corresponding figures or in the source data file linked to each figure. Throughout the revised manuscript, we have also systematically specified the CMV-serotype of each donor when determined. This was done either by clearly indicating the CMV-serotypes in the figure labels or legends, or by providing a color-code for CMVneg, CMVpos and Not Determined. Healthy donor CMV serotypes are also provided as a source data file. These clarifications confirm that healthy donor CMV serostatus does not affect DOT cell expansion, phenotype and anti-CMV reactivity.

Of note, CMV serology must be specifically requested from our local blood bank. For this reason, it was not systematically determined for all donors in the later phases of the project, after having established that CMV serostatus does not impact DOT cell expansion, phenotype and anti-CMV reactivity.

2. In the materials and methods it is mentioned that, unless stated otherwise, IL-18 is used in the co-culture experiments. This creates confusion when going back and forth between main figures and supplementals, with the need to constantly re-read the legends searching for a mentions of IL-18 absence before being able to compare figures. In our opinion, adding a simple 'IL-18+' or 'IL18-' label in the figures would improve readability and interpretability.

Reply: We have now clearly indicated in the legends of Figures 2, 3, and 5 which cocultures were performed in the presence of 50 ng/mL IL-18. We have additionally added a +IL-18 label in these Figures.

3. Many abbreviations and acronyms are used, which are not consistently explained (sometimes in the main text, sometimes only in the figure legend). We urge the authors to critically go through the main text and explain abbreviations/acronyms appropriately. Highlighting (bold letters) and using a glossary would increase readability, if this is allowed in the Nature Comms format.

Reply: We carefully checked and defined all abbreviations. However, Nature Communications does not allow to highlight abbreviations and does not recommend to provide a glossary.

4. The figure panels (especially the flow cytometry gates panels) are too small to read comfortably from an A4 printed paper format. We strongly advise to consider using larger panels.

Reply: Following the reviewer's recommendation, we enlarged Figure panels, especially the flow cytometry gate panels in **Figure 1, Supplementary Figure 1 and Supplementary Figure 2.**

5. The authors address the risks of off-target reactivity in the discussion. What is the expected side effect profile in humans, this should be considered. More data are needed, not only mock infected cell lines (which also do not induce 'zero' response) and a reference to an ongoing clinical study in leukemia patients. The SOT patient with an ongoing infection (activated stage, possibly other viruses) and the TCR independent mechanism as shown in this manuscript should be taken into account and when possible other targets should be included in the experiments.

Reply: In the initial studies describing the generation of DOT cells (PMID: 27307596; PMID: 30894378), their absence of reactivity was shown against healthy alpha-beta T cells, B cells, NK cells and other leukocyte populations. In addition, in our study, we show minimal DOT cell reactivity against healthy primary fibroblasts and endothelial cells. We believe that it is already quite a good demonstration that DOT cells are at low risk of off-target reactivity. We do not think that adding other targets will add much to the demonstration. As a matter of fact, DOT cells have been approved in a clinical trial (NCT05886491), which will be a more relevant way to assess the safety of this product. This important point mentioned by the reviewer is addressed in the discussion of the revised manuscript.

Specific considerations:

6. Methods section: no mentioning of off target scores of the used sgRNAs for the KO model. Was this check performed? How else have off-target editing effects been excluded? Please explain.

Reply: Off-target scores of gRNAs have indeed been evaluated. gRNA were designed using the CRISPOR algorithm (described in Haeussler et al. Genome Biology 2016, DOI: <https://doi.org/10.1186/s13059-016-1012-2>) using specificity as the main criterion. This specificity was assessed: (i) by the CFD specificity scores (Doench et al., Nature Biotechnology 2016, doi.org/10.1038/nbt.3437; Tycko et al., Nature Communications

2019, doi: 10.1038/s41467-019-11955-7) which is an accurate score ranging from 0 to 100, with 100 being the highest specificity and (ii) by looking closely at the potential number of off-target exons.

Characteristics of each gRNA used in this study according to CRISPOR:

Gene –gRNA	CFD specificity score	Number of exonic sequences with 3 mismatches	Number of exonic sequences with 4 mismatches
TDRC1-gRNA1	95	0	5
TDRC1-gRNA2	92	0	4
ITGAL1-gRNA1	93	0	3
ITGAL1-gRNA2	93	1	4

Furthermore, all the experiments were carried out using a high specificity Cas9 protein from IDT : Alt-R™ S.p. HiFi Cas9 Nuclease V3, which has been engineered to dramatically reduce off-target effects (Vakulskas et al, Nat. Med. 2018, doi: 10.1038/s41591-018-0137-0).

This information has been added to the Methods section (Section: CRISPR-Cas9 knockout in DOT cells).

7. Line 375: Please elaborate on how these donors are selected and is this enough to make the conclusion in line 390-391?

The reviewer refers to the following conclusion: “Finally, we observed no impact of the CMV serotype in this phenotypic analysis, as illustrated for 3 CMVneg and 3 CMVpos donors”.

Reply: No specific selection criteria were applied to include these donors, apart from their CMV-serotype. To further support the conclusions of these experiments, we performed a statistical comparison between CMVneg and CMVpos donors after DOT cell expansion. These analyses have now been added to the revised manuscript as **Supplementary Figure 3**. We compared either the percentage of positive cells for each marker (**Fig. S3A-E**), or the median fluorescence intensity of DNAM-1, NKG2D and LFA-1 (for which >90% of Vδ2^{neg} γδ T cells are positive after expansion, **Fig. S3F**). These analyses confirm that the CMV serotype does not significantly impact DOT cell phenotypes after expansion. Of note, as KIRs were not expressed by DOT cells after expansion, no statistical analysis comparing CMVneg and CMVpos was performed for these markers.

8. Figure 1E: include CMV status (donor 1-2 = CMV neg, donor 3-4 = CMV pos)

Reply: The CMV status of the donors is now indicated. In addition, to strengthen these data, we phenotyped 2 additional CMVpos donors. **Figure 1E** now includes 3 CMVneg donors and 3 CMVpos donors. For each cell population, the percentages are not

significantly different between CMVneg and CMVpos donors (unpaired t test, P values described in the results section).

9. Figure 1F: In the text the terms central memory or effector cells are used, in the legend within the figure this is translated to CD27 and CD45RA status. Provide a label in the figure for the central memory or effector population as well for clarity. Also, SD are shown, but no statistical analyses are provided.

Reply: A label for each population has been added in **Figure 1F** and the legend has been clarified to indicate naïve: CD45RA⁺ CD27⁺; T central memory (TCM): CD45RA⁻ CD27⁺; T effector memory (TEM): CD45RA⁻ CD27⁻; T effector memory re-expressing CD45RA (TEMRA): CD45RA⁺ CD27⁻. Statistical analyses (unpaired t test for each population) are now provided and described in the figure legend of the revised manuscript.

10. Figure 1F is associated with figure 5E (where a similar analysis is performed in the murine CMV in vivo model). We suggest to model the 2 figures in a similar way so they can be compared side by side and similarities in orthologous cell populations are immediately apparent.

Reply: **Figure 1F** and **Figure 4E** of the revised manuscript (previously Figure 5E) are now presented identically following the reviewer's suggestion. In particular, naïve, TCM and TEM populations are clearly labelled in both figures. A statistical analysis was also added to **Figure 4E** to mirror **Figure 1F**.

11. Line 409-410: more elaborate explanation of why authors chose to focus on IFN- γ is needed. They refer to a paper, but a short explanation of why this is the most interesting factor (compared to the other targets in the CMV-induced secretome) would be appreciated.

Reply: Since its initial description as a soluble macromolecule “interfering” with viral replication (PMID: 17533668), IFN γ has been extensively characterized as a pleiotropic cytokine that modulates both innate and adaptive antiviral immune responses and can also directly inhibit virus replication (as reviewed for instance in PMID: 30402328). In previous studies from our group, IFN γ has consistently been the best read-out of $\gamma\delta$ T cell reactivity against CMV-infected cells (PMID:22180442; PMID: 3262235; PMID: 34725108) and we have shown that the ability of activated $\gamma\delta$ T cell supernatants to control virus replication is mediated by IFN γ (PMID: 22180442). These clarifications have been added to the Results section ‘DOT cells display anti-CMV effector functions’.

12. Figure S3A: contains endothelial cells, effect of CMV infection is minimal. Please comment on why this is (and also why in S3D they don't seem to react to CMV induction without IL-18, but with IL-18, suddenly IFN-g is increased).

Reply: Infection of endothelial cells by CMV *in vitro* has been described to be far less efficient than for fibroblasts, when the CMV strain used is propagated in fibroblasts as

performed in our study (see for instance PMID: 2550582). The low percentage of infected endothelial cells shown in **Figure S4A** of the revised manuscript was therefore expected. As a consequence, the reactivity of DOT cells is expected to be less important against infected endothelial cells compared to infected fibroblasts, as shown in **Figure S4D** of the revised manuscript. When IL-18 is added to co-cultures (**Fig. 2B**), it potentiates CMV-induced IFN γ production (as described in Guerville et al., Oncoimmunol 2015; PMID: 26155394). In cocultures with endothelial cells, IL-18 therefore reveals a reactivity otherwise invisible.

13. Figure S3B: FMO of infected cells is missing. Can the addition of these cells lead to an increase in the CD107a fluorescence channel? These data should be added.

Reply: We would like to clarify here that the CD107a stainings were performed on DOT cells, not on infected or uninfected target cells. In the revised manuscript, a label was added in **Figure S4B** (previously S3B) indicating “gated on V δ 2^{neg} $\gamma\delta$ T cells” to clarify. If the reviewer meant the FMO of DOT cells cultured with infected or uninfected cells, those are similar as shown below (**Figure R1**). The addition of target cells (mock or CMV-infected) to DOT cells does not induce any detectable change in their size or autofluorescence, and therefore in the % of cells in the CD107a-positive gate in FMO controls.

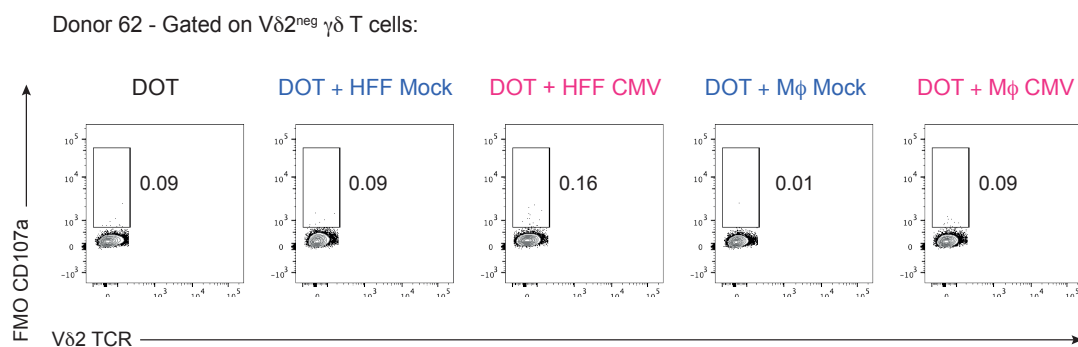


Figure R1: Related to **Figure S4B**. Fluorescence Minus One (FMO) controls are shown from one representative experiment (DOT donor 62) in the different coculture conditions. Dot plots were gated on V δ 2^{neg} $\gamma\delta$ T cells.

14. Line 412: IL-18 is used in most co-cultures (please add in the figure panel e.g. in S3D when it is used and when not), but how does this relate to an expected effect in vivo in a human? As it seems IL-18 is needed to increase functionality of the gd T-cells. Would you expect to see the same effects as observed in vitro, when in an in vivo situation you cannot co-stimulate with IL-18? Please comment on that.

Reply: We have now clarified when IL-18 is used in the different figures (see response to comment 2 above). Regarding the expected effect in patients, IL-18 has been detected at high level in the sera of renal transplant recipients during CMV-infection, concomitantly with elevated levels of IFN γ (PMID: 20632887). In an experimental model of CMV infection in mice, IL-18 has been shown to be produced by dendritic cells in infected spleens (PMID: 39134803). In addition, we have previously shown that IL-18 can be produced by CMV-infected human venous endothelial cells (HUVEC, PMID: 26155394). It is therefore reasonable to expect that IL-18 produced *in vivo* will help DOT cells to control CMV. These comments have been included in the discussion section of the revised manuscript.

15. Figure S3C: no statistical analyses. Different secretome factors are shown (with various efficiency of induction by CMV infection). IFN- γ was selected as the most important one. Was this based on statistical significance?

Reply: In agreement with the reviewer's comment, we have added a statistical analysis on the results presented in **Figure S4C** (previously S3C), showing that both IFN γ and GM-CSF are significantly induced when DOT cells were co-cultured with CMV-infected cells (compared to DOT cells cultured alone). This induction was particularly striking for IFN γ after coculture with CMV-infected macrophages. Because of the well-established function of IFN γ against viral replication (see response to comment 11), by contrast to GM-CSF, we then focused on IFN γ .

16. Figure 2D: please explain the use of HSV and VZV in more detail. Why are they used as controls? Are there controls used to show successful infection, what is their infection rate? Infection and expression of specific antigen (IE-1) is shown for CMV-infected cells, but not for HSV and VZV-infected cells. Is this the correct infection model these viruses? A bit of background is needed. And also are these still considered the right controls considering that later in the paper is shown that the cells do not require TCR engagement.

Reply: HSV and VZV are used as controls because they belong to the same family of Herpesviridae as CMV and share genetic similarities (PMID: 2552898). They also can be easily compared to CMV *in vitro* since they can readily infect fibroblasts, which are the prototypic target cells used for *in vitro* infection by the three viruses. We have already shown in previously published studies (Halary et al., JEM 2005 PMID: 15897274; Kaminski et al., JID 2021 PMID: 32622351) that HSV-1 and VZV-infected fibroblasts do not activate V δ 2^{neg} $\gamma\delta$ T cells. This is consistent with the absence of $\gamma\delta$ T cell responses in kidney transplant recipients infected with HSV or VZV, by contrast to patients infected with CMV (Dechanet et al., JID 1999 PMID: 9841815; Kaminski et al., JID 2021 PMID: 32622351). *In vitro*, successful infection can be determined by observing the cytopathic effects of these 3 lytic viruses by microscopy. We then used cultures with the same rates of cytopathic effect (in between 50% and 100%) to compare DOT cell reactivities.

Interestingly, and as mentioned by the reviewer, even though the activation of DOT cells by CMV-infected cells is TCR-independent, it remains specific of CMV when compared to VZV and HSV, indicating that the induction of other molecules than a TCR ligand can be

specific of one virus. Alternatively, VZV and HSV could encode unique immune evasion mechanisms absent during CMV infection.

17. Figure 2 figure legend ‘Mock-infected fibroblasts were used to set the GFP-positive gate’: please show the gating used in the supplements.

Reply: The gating strategy used to obtain these results is now shown in **Figure S4E** and mentioned in the results section of the revised manuscript.

18. Line 439-441: ‘addition of a V δ 1 TCR blocking...’ Could this be because the effect is mainly through V δ 3 then? Please comment.

Reply: We agree with the reviewer’s comment: the use of an anti-V δ 1 blocking antibody does not preclude a remaining response by V δ 3 T cells, even though they represent a minor population in the DOT cell product, as shown in **Figure 1E**. However, TCR knockout experiments (shown in **Figure 3C** and **D** and **Figure S5** of the revised manuscript) in which the TCR δ constant region shared by all $\gamma\delta$ T cells is targeted, clearly showed a $\gamma\delta$ TCR-independent IFN γ secretion in response to CMV.

19. Line 443: ‘two distinct donors’ – why are these donors selected and what makes them ‘distinct’ (in what respect? CMV serology?). What is their CMV Serological status?

Reply: As mentioned above, donors were not selected based on specific characteristics but were freshly amplified specifically to perform these knockout experiment. The term “distinct” was used to indicate that the knockout experiments were performed on two independent donors. To avoid ambiguity, we have also replaced “distinct” with “independent” when referring to donors in the revised manuscript. Additionally, the CMV serological status of the three donors used for TCR knockout experiments is now indicated in the **Figure 3** legend of the revised manuscript.

20. Line 446: ‘edition’ $\diamond$ is ‘editing’ meant here?

Reply: ‘Edition’ has now been replaced by ‘editing’ in the text.

21. Line 450: explain briefly what the expected effect is of PMA ionomycin (used as a general inductor, this may be common knowledge in the field but requires a short explanation for the non-hematological experts reading this paper.

Reply: Stimulation with PMA and ionomycin is a commonly used positive control to induce a polyclonal T cell activation. PMA notably induces protein kinase C activation, while ionomycin is a calcium ionophore triggering calcium-dependent signaling pathways. Altogether, stimulation with these two compounds bypasses the TCR complex to activate several intracellular signaling pathways. We have added a summary of these explanations in the revised manuscript (Results section “DOT cells do not need their TCR to recognize CMV-infected cells”).

22. Line 452: remanent = remnant or better even 'remaining' ?

[Reply:](#) The correction has been made in the text.

23. Figure 3A: Here fold increase is depicted while in the rest of the figure pg/ml IFN γ production is depicted. It makes it easier to read these figures when the axes are comparable

[Reply:](#) We chose to plot fold increases in **Figure 3A** because the basal production of IFN γ by DOT cells (in the absence of mock-infected or CMV-infected fibroblasts) can vary greatly from one donor to another, which was particularly visible in this experiment. To illustrate this, please find below the corresponding raw data, where IFN γ production is shown in pg/ml (**Figure R2**). This important variation makes it difficult to compare “transwell” and “contact” in the conditions where DOT cells are co-cultured with CMV-infected fibroblasts. For this reason, we propose to keep fold increases in **Figure 3A**.

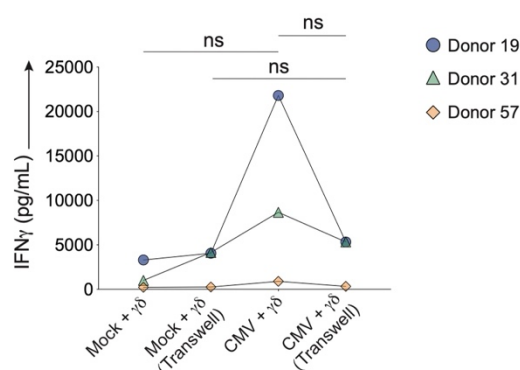


Figure R2 (related to Figure 3A). DOT cells were cultured alone or in the presence of mock or CMV-infected fibroblasts, either in contact or separated by a Transwell system, in the presence of IL-18. IFN γ secretion is shown for three experiments performed on independent donors. Each datapoint represents the mean of culture triplicates. ns $P > 0.05$, paired t-test.

24. Figure 3B: lacks a positive control for the blocking effect of $\alpha V\delta 1$

[Reply:](#) We thank the reviewer for this important comment. As a positive control for assessing the blocking efficacy of the V $\delta 1$ antibody, we stimulated DOT cells obtained from four independent donors with a plate-bound anti-CD3 antibody (clone OKT-3), which is known to robustly induce IFN- γ secretion in these cells. Prior to stimulation, DOT cells were pre-incubated with either an isotype control antibody or the V $\delta 1$ -blocking antibody. Pre-treatment with the V $\delta 1$ -blocking antibody resulted in a significant reduction in IFN- γ production compared to the isotype control. These results are

presented in **Figure R3** below. Additionally, to strengthen the results shown in **Figure 3B**, we repeated the experiment, increasing the sample size to $n = 8$ donors.

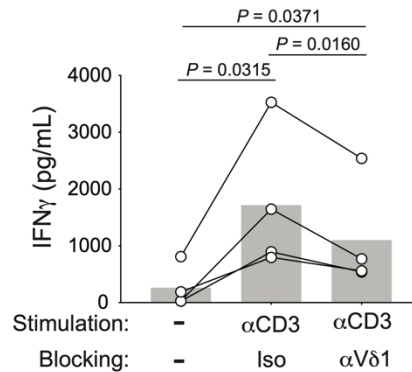


Figure R3: Assessment of the blocking efficacy of the V δ 1 antibody.

DOT cells were pre-incubated with either an isotype control antibody or an anti-V δ 1 blocking antibody prior to stimulation with plate-bound anti-CD3 (clone OKT-3) in the presence of IL-18 for 24h. IFN γ secretion was measured. Each datapoint represents the mean of technical triplicates from an independent donor ($n = 4$). Statistical comparisons were performed using ratio paired t tests.

25. Figure 3D (and also in other figure panels where similar antibodies are used): it is unclear whether antibodies used are inhibitory or inducing. Please improve the colour scheme in a way to clearly show the expected effect, so it is clearly visible in the figure.

Reply: We have now directly indicated in **Fig. 3B**, **Fig. 3E**, **Fig. 3F**, **Fig. 3H** and **Fig. S5A (previously Figure 3D)** whether the antibodies used were activating or blocking antibodies. This information is now also indicated in the corresponding figure legends.

26. Figure 3G: ‘representative’ donor – only 1 donor is shown so this is an $N=1$ figure, while donor variability can be important here.

Reply: We thank the reviewer for this important comment. We fully agree that donor variability is a critical consideration. To address this, we repeated this experiment on three additional donors and now show data from $n=6$ donors in total. In the revised **Figure 3E** (previously Figure 3G) one dot corresponds to the average of culture triplicates from one donor. In this expanded dataset the same conclusion can be drawn as in the initial data set: blocking antibodies directed against NKp30, NKp44, NKG2D, or DNAM-1 do not significantly reduce IFN γ production by DOT cells in response to CMV-infected fibroblasts, as compared to isotype control conditions. These results show that these activating NK receptors are not individually required for CMV-induced IFN γ production in DOT cells.

27. Line 472: it is not entirely convincing from the presented data that the main response is through LFA-1. We do not agree with the word 'mainly', difference between ICAM and LFA-1 is not convincing.

Reply: As suggested, we have deleted the word 'mainly' and rephrased this sentence. In the revised manuscript, we have now written: "Importantly, in coculture with CMV-infected fibroblasts, blockade of LFA-1 or ICAM-1 systematically reduced IFN γ secretion by DOT cells expanded from six independent donors, with the strongest effect observed when both were blocked simultaneously".

28. Line 475: edition = editing?

Reply: 'Edition' has now been replaced by 'editing' in the text.

29. Figure 4D+E(bottom): because the KO of donor 2 is less efficient, there is a large spread in IFN induction in donor 2 and ns difference between mock and CMV cocultured cells, so we are effectively looking at data from only 1 donor that is conclusive. Based on an N=1 we cannot draw these strong conclusions. Additional data is needed here.

Reply: We appreciate this important comment and fully agree that conclusions drawn from a single donor with efficient knockout are insufficient. To address this comment, we attempted LFA-1 knockout in DOT cells from three additional donors. Knockout efficiency was assessed by flow cytometry (**Figure R4**).

Unfortunately, these three attempts to obtain a full knockout failed (as shown in **Figure R4 below**), as LFA-1 expression remained high following editing. We also attempted to enrich LFA1-negative cells by cell sorting, but in all cases, we were unable to recover a sufficient number of viable cells for downstream functional assays. Given these technical limitations, we have decided to remove the initial LFA-1 knockout data from the manuscript. Instead, conclusions about the involvement of LFA-1 in CMV-specific reactivity are now based on functional experiments using anti ICAM1 and LFA-1 blocking antibodies, now performed in six independent DOT donors in total. These results consistently support a role for LFA-1 in DOT cell anti-CMV reactivity.

We have revised **Figure 3** and **Figure 4** accordingly. The two figures are now merged in one. In the revised manuscript, we have mentioned this technical limitation in the Results section. We have also revised our interpretation and discussion of this mechanism accordingly (in the Discussion section).

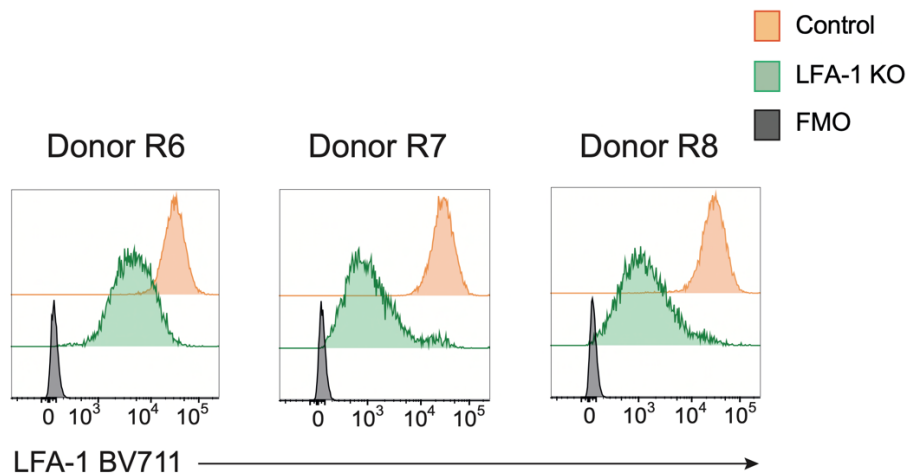


Figure R4: DOT cells from three independent donors (donors R6, R7 and R8) were knocked out for LFA-1 (ITGAL gene) and KO efficiency was verified by flow cytometry. LFA-1 staining is shown for control (orange), LFA-1 knockout (green) and FMO (grey) conditions.

30. Line 481 - 482: Not enough data here to conclude it is a key-mediator (see comment above).

Reply: We agree with the reviewer. While blocking experiments (now confirmed in three additional donors) support a role for LFA-1, we were unable to validate this through LFA-1 knockout due to technical limitations. Accordingly, we have revised the manuscript text to adopt a more cautious interpretation: “Altogether, stimulation and blocking experiments suggest that LFA-1 contributes to DOT cell anti-CMV reactivity”.

31. Figure 5E: see remark for Fig 1E. Mirror these 2 so you can compare human and mouse.

Reply: **Figure 1F** and **Figure 4E** (previously Figure 5E) of the revised manuscript are now presented identically.

32. Figure 5F: conclusion on LFA-1 cannot be drawn. For human, MdFI is measured in figure 1H to show LFA-1 is induced (in heat map this is comparable for day 0 and day 21. In 5F there is no remarkable difference in LFA-1 between day 0 and day 7. We cannot assume without the MdFI that the effect is the same as in human, so it cannot be said that (line 507) LFA-1 expression is upregulated.

Reply: In order to provide a more systematic statistical analysis of the data presented in **Fig. 4F** (previously Fig. 5F), we compared the % of mouse $\gamma\delta$ T cells positive for each marker (**Fig. S6A** of the revised manuscript) as well as the MdFI of each marker on $\gamma\delta$ T

cells (**Fig. S6B** of the revised manuscript). Notably, the comparisons of LFA-1 % and MdFI at day 0 and day 7 show a significant increase. These analyses have now been added to the revised manuscript as **Supplementary Figure 6**, to support the following conclusions “Phenotyping of murine $\gamma\delta$ T cells before and after amplification showed an increase in activation (CD25), a lack of immune-checkpoint inhibitors such as PD-1 or TIM-3, and an upregulation of NK activating receptors (DNAM-1, NKG2D), cytotoxicity receptors (FasL and TRAIL), as well as LFA-1 expression, independently of previous CMV priming”.

33. Figure 5G: 2 doses of cells given $\diamond$ how does this compare to the dosing regimen in human? Also: unclear label, in the top it says C57BL/6J +/- MCMV, is that the serological status? Because below it says ‘MCMV i.p.’ which is infection time point. The top reads as if the mice are administered the cell therapy and simultaneously receive the MCMV virus or not. Please change to less confusing label.

Reply: Although a $\gamma\delta$ T cell therapy has not yet been tested in humans to treat CMV infection, we can compare the numbers of mouse $\gamma\delta$ T cells administered in our in vivo experiments (**Figure 4** of the revised manuscript) to the numbers of infused CMV-specific CD8⁺ T cells in Smith et al. (Clin Infect Dis 2019 DOI: 10.1093/cid/ciy549) and Khoury et al. (Am J Transplant 2024, DOI: 10.1016/j.ajt.2024.04.009). These two pioneering studies evaluated CMV-specific CD8⁺ T cell adoptive transfer to treat CMV infection in solid-organ transplant recipients.

In Khoury et al., patients received a median of two infusions of virus-specific T cells at a dose of $50 \times 10^6/\text{m}^2$ body surface area. In Smith et al., patients received up to six doses of in vitro-expanded T cells at $10\text{--}20 \times 10^6/\text{m}^2$ every two weeks.

In our experiments, Rag^{-/-}γc^{-/-} mice received 10^6 $\gamma\delta$ T cells on day 0 and day 7. Given their average body surface area of 0.007 m^2 , this equates to $143 \times 10^6/\text{m}^2$ per dose. This means the mice received approximately three times the maximal dose used in Khoury et al., and 2.4 times the maximal dose used in Smith et al. Thus, the $\gamma\delta$ T cell dose administered to the mice falls within the range of those used in these two human trials.

We have also modified the labels and legends in **Fig. 4G** of the revised manuscript (previously Fig. 5G) to improve clarity.

34. Figure 5H: the lethality differences are not very convincing. Plus there seems to be a difference between MCMVpos and neg (this is ns, but the numbers per group are different), any comments on this?

Reply: In the first version of the manuscript, the differences in mouse numbers between groups resulted from pooling three independent experiments, each with varying numbers of mice. To strengthen our findings and increase group sizes, we conducted an additional experiment and incorporated the results into the existing dataset (**Figure 4H** of the revised manuscript, previously Fig.5H). We now present data from 13 Rag^{-/-}γc^{-/-} mice that received $\gamma\delta$ T cells from MCMVpos C57BL/6J mice (+7), 22 mice that received $\gamma\delta$ T

cells from MCMVneg C57BL/6J mice (+5), and 18 control mice (+3). Our results confirm that the transfer of $\gamma\delta$ T cells from both MCMVpos and MCMVneg mice significantly improved the survival of MCMV-infected Rag-/- γ c-/- mice compared to the RPMI control ($P < 0.0001$ and $P = 0.0001$, respectively; log-rank Mantel-Cox test), with no significant difference between MCMV status of donor mice ($P = 0.0837$).

35. Figure 5I: this is a confusing experiment. It tells us that in mice that are still alive, the viral load in the organs is lower than in mice that were lethally infected. The mock treated mice (with RPMI) all had a high titer in the organs and were lethally infected. But of the treated mice, some still end up in the lethally infected groups with high viral load. Ergo, there seems to be a trend but you cannot draw a causative conclusion that the lower viral load is because of the cell therapy.

Reply: We appreciate this insightful comment. We agree that Fig. 5I shows a correlation (mice that survived MCMV infection had lower viral loads in the spleen, liver, and lungs compared to those that succumbed) but does not allow to draw a causative conclusion. In the revised manuscript this figure is now positioned as **Fig. S7A**.

To formally test whether $\gamma\delta$ T cell adoptive transfer can control MCMV viral load in infected mice, we performed an additional experiment using the same protocol as in the previous Fig. 5I. MCMV-infected Rag-/- γ c-/- mice were adoptively transferred with $\gamma\delta$ T cells amplified from MCMVpos ($n=5$) or MCMVneg ($n=5$) C57BL/6J donor mice, or with RPMI as a control ($n=5$). All mice were sacrificed at day 14 post MCMV infection to allow direct comparison of viral loads. This experiment showed that the transfer of $\gamma\delta$ T cells amplified from both MCMVpos and MCMVneg donor mice significantly reduced viral loads in the spleen, liver, and lungs compared to RPMI-treated controls, demonstrating that $\gamma\delta$ T cell adoptive transfer induces a decrease in viral load. This new experiment is now included in the revised manuscript as **Figure 4J**.

36. Figure 5J: Please comment on what the ‘effector memory population’ would be. In the text it is said the transferred T cells had a ‘TEM phenotype’ (line 521-522) but TEM is characterized as CD62L⁻, CD44⁻ in the figure legend. The majority of cells is CD62L⁻, CD44⁺.

Reply: Thank you for this valuable comment. There was an error in the legend of this figure: TEM cells are CD62L⁻ CD44⁺, not CD44⁻. We have corrected the legend of **Figure 4E** (previously Figure 5E) accordingly.

37. Line 524-525: this is a strong conclusion which is not entirely supported by the presented data mainly in figure 5H and I. In addition, it should be discussed whether this is the right mouse model to make these conclusions.

Reply: Please refer to our responses to comments 34 and 35, where new experiments have been carried out to support these conclusions. We have also revised our conclusion

as follows: “Together, these findings demonstrate that adoptively transferred $\gamma\delta$ T cells can migrate to MCMV-infected organs, control viral replication, and protect immunodeficient mice from lethal MCMV infection”.

The main advantage of this mouse model of MCMV infection is its similarity to human CMV pathogenesis. It induces a systemic disease, as observed in humans, and is widely used, making it an appropriate model for testing new cell therapies. However, its primary limitation is that it does not allow testing of human DOT cells. Nevertheless, we addressed this limitation by generating murine “DOT-like” cells that exhibit the same central memory phenotype as human DOT cells and shared their anti-CMV activity. In the future, humanized CMV mouse models could provide additional insight into the efficacy of human DOT cells *in vivo*.

These additional points have been incorporated into the discussion section of the revised manuscript.

38. Title of last results paragraph (p24): we do not agree that the DOT cells ‘meet cell therapy requirements’. This is a generalization. While the DOT cells may meet some requirements, the data presented does not unequivocally show it meets all the cell therapy requirements.

Reply: This is a fair comment. To take it into account, we have split the final section of the Results into two parts with more appropriate titles.

- The first paragraph, which includes **Table 1** and **Fig. 5A-D**, is now titled “Autologous CMV-reactive DOT cells can be obtained from SOTR”.

- The second paragraph, which includes **Fig. 5E-I** and **Fig. S8**, is now titled “DOT cells remain functional under immunosuppression”.

39. Line 544: please explain why it is necessary to add IL-15 to these cultures. Is IL-15 needed to make the cells less vulnerable for the immune suppressive drugs? And how does this relate to the *in vivo* situation?

Reply: IL-15 is essential to support the survival of DOT cells both *in vitro* and *in vivo* (in NSG mice), as demonstrated in previous studies (Mensurado et al., Blood 2024 PMID: 38437507; Blanco-Dominguez et al., Nat Cancer 2025 PMID: 40240620). Following expansion, DOT cells are systematically rested in complete medium supplemented with IL-15 prior to their use in functional assays, including the proliferation and reactivity assays in the presence of immunosuppressive drugs shown in **Figure 5** of the revised manuscript (described in the Methods section).

Regarding the *in vivo* relevance, several studies have shown that IL-15 levels increase during CMV infection. For instance, increased IL-15 mRNA levels have been observed in the bone marrow of CMV-seropositive individuals compared to CMV-seronegative individuals (Pangrazzi et al., Front Immunol 2017, PMID: 28674537). Additionally, ICU patients exhibit higher levels of IL-15 before CMV reactivation, compared to controls

(Chiche et al., Crit Care Med 2012, PMID: 22971588). Based on this evidence, we can speculate that the production of IL-15 during CMV infection may similarly support DOT cell survival *in vivo*.

40. Figure 6F and H: please comment on the very low proliferation index observed (high contrast with the DOT expansion protocol from healthy donors) and the relatively low amounts of live cells? H also doesn't have statistics.

Reply: The aim of this experiment was to evaluate the survival and proliferation of DOT cells *in vitro*, in the presence of immunosuppressive drugs. For this, we measured their proliferation capacity and viability in culture during 15 days, **after** the end of their amplification. Once amplified, their proliferation index was expected to be lower than during their amplification phase.

The number of living cells did not exceed 50% at the end of the culture, indicating a certain degree of cell death. Nevertheless, their proliferation capacity still allowed for the renewal of a sufficient number of DOT cells to maintain the DOT cell population at day 15 at the same level as on day 0 (see **Fig. 5G** of the revised manuscript).

Furthermore, we would like to highlight that for the experiments presented in panels E-F-G-H-I, we used DOT cells amplified from healthy donors.

In summary, their ability to persist for 15 days *in vitro* suggests that DOT cells could survive *in vivo* for days or weeks after infusion. This was confirmed in our mouse model, where *in vitro* amplified $\gamma\delta$ T cells were found in infected organs more than 15 days after infusion.

Fig. 5H now shows P values (determined by repeated-measures one-way ANOVA).

41. Figure 6I: what is the dotted line representing? Mock+DOT as a baseline? Please explain?

Reply: The dotted line indeed represented Mock+DOT at baseline but did not provide any relevant additional information to interpret this figure. As a result, we have removed the dotted line to avoid any confusion in the interpretation of the figure.

42. Figure legend for 6 on p. 44. Inconsistent representation of 'donors' vs. 'experiments'. Sometimes donors are mentioned, sometimes it's 'representative experiments' but are these then from same donor? Please make this consistent and clear how many donors (patients or healthy) are used per experiment.

Reply: We thank the reviewer for pointing this out. We have now clarified the number of donors used in each panel of the figure. Specifically, for panels A to D, DOT cells were amplified from 8 kidney transplant recipients. For panels E to I, DOT cells were amplified

from healthy donors. 4 donors were used for panels E and F, 3 donors for panels G and H, and 5 donors for panel I.

In the revised manuscript, Figure 6 has been renumbered as **Figure 5**.

Reviewer #2 (Remarks to the Author):

1. The authors present interesting data regarding an adoptive gamma-delta T cell based approach using only 50 ml of patient blood to tackle CMV disease which are of significance to the field, however to increase rigor, the amount of donors should be increased to show be more convincing.

Reply: We thank the reviewer for this insightful comment. In response, we have increased the number of donors analyzed to strengthen the robustness of our findings. Specifically:

- **Fig. 1E:** this analysis is now performed on DOT cells from n=6 donors (three CMVneg and three CMVpos) instead of n=4 previously.
- **Fig. 3B:** this analysis is now performed on DOT cells from n=8 donors (previously n=4).
- **Fig. 3C and 3D:** $\gamma\delta$ TCR knockout experiments were performed on an additional DOT donor, for a total of n=3.
- **Fig. 3E:** we now present an analysis on n=6 donors, instead of one representative donor shown previously.
- **Fig. 3H:** ICAM1 and LFA1 blocking experiments were performed on three additional DOT donors, for a total of n=6.
- **Fig. 5A-D:** DOT cells were amplified from three additional kidney transplant recipients, for a total of n=8. In **Fig. 5D**, all 8 patients are now represented in the same graph (each datapoint corresponds to the mean of culture triplicates for one patient).

Across the revised manuscript, each analysis is now based on a minimum of n=3 healthy donors or patients. Technical replicates are no longer represented in figure panels, with the exception of **Fig. 3D and Fig. S5** ($\gamma\delta$ TCR knockout experiments), where each donor is shown on an individual graph.

2. I would recommend the following adjustments:

- l. 370: The numbers/percentages of TEMRA should be mentioned.

Reply: We have now included the relevant percentages in the revised Results section:

- In CMV-seronegative donors, naïve cells represented $74.0 \pm 18.8\%$ of $V\delta 2^{\text{neg}}$ $\gamma\delta$ T cells at day 0.
- In CMV-seropositive donors, TEMRA cells represented $41.2 \pm 19.8\%$ of $V\delta 2^{\text{neg}}$ $\gamma\delta$ T cells at day 0.

- At day 21, DOT cells from both CMV-seronegative and CMV-seropositive donors were predominantly central memory cells ($88.1 \pm 9.2\%$ and $88.1 \pm 9.6\%$ of $V\delta 2^{\text{neg}}$ $\gamma\delta$ T cells respectively).

3. - At least 6 donors should be shown for each experiment, preferentially 3 females and 3 males, technical replicates are less relevant, they can be used to calculate a mean/median for each donor

Reply: We appreciate the reviewer's suggestion to strengthen the robustness of our study by increasing donor numbers and considering sex balance.

While we were not able to include six donors for every experiment, we have increased donor numbers throughout the manuscript wherever possible (as detailed in our response to comment 1). All analyses are now performed on a minimum of three independent donors, and all analyses presented in Figure 1 and 2 were performed on a minimum of 6 donors. Regarding donor sex, we regret that we do not have access to this information for the healthy donor samples used in this study. Regarding kidney transplant recipients, the sex of the patients is now included in Table 1 of the revised manuscript.

We agree with the reviewer that technical replicates are less informative than biological replicates, and we have revised the presentation of our data accordingly: individual data points now represent means for each donor, and technical replicates are no longer shown in the figures (except in **Fig. 3D** and **Fig. S5**, where each donor is presented separately).

We hope that this expanded dataset addresses the reviewer's concerns, as it enhances the robustness of our findings.

4. The appropriateness of the mouse experiment should be further discussed regarding differences of humans and mice gamma-delta T cells.

Reply: The main advantage of this mouse model of MCMV infection is its similarity to human CMV pathogenesis. It induces a systemic disease, as observed in humans, and is widely used, making it an appropriate model for testing new cell therapies. Furthermore we have previously shown that MCMV induces long-term expansion of murine $\gamma\delta$ T cells that protect mice from the infection, very similarly to what can be observed in human CMV infection (PMID: 25747674 ; PMID: 38976755). MCMV is then precisely an interesting model to make parallels between human and mouse $\gamma\delta$ T cells. However, the primary limitation of this MCMV model is that it does not allow testing of human DOT cells. Nevertheless, we addressed this limitation by generating murine "DOT-like" cells that exhibited the same central memory phenotype as human DOT cells and shared their anti-CMV activity. In the future, humanized CMV mouse models could provide additional insight into the efficacy of human DOT cells in vivo. We have incorporated these comments into the discussion section.

5. l. 501: All data should be shown either in the supplement or main figures, not showing data is not acceptable.

Reply: We agree with the reviewer that all data described in the manuscript should be shown. In order to simplify the lecture for the reader, we have deleted this sentence from the manuscript.

6. l. 528: Is there supporting evidence for this claim, especially having in mind the TRACE study?

Reply: Both autologous and allogeneic virus-specific cell therapies have been or are currently being tested in phase I trials in SOTR (PMID: 29982441; PMID: 38643944; ACTRN12621000323820) or following allogeneic stem cell transplantation (TRACE study, NCT04832607). We agree that there is currently no direct evidence that “an allogeneic T-cell therapy would eventually be rejected by the host immune system, thus limiting its long-term efficacy”. This was merely a hypothesis, and we have removed this sentence.

7. l.532: Actually, there are many reports showing successful expansion of autologous T cell products from kidney transplanted patients, please show evidence or adjust the statement (e.g. 10.3389/fimmu.2019.01148).

Reply: We thank the reviewer for this fair comment, but there is currently no data showing successful expansion of autologous $\gamma\delta$ T cells from solid organ transplant recipients receiving immunosuppressants. To address this comment, we have adjusted the introduction of this paragraph and added references regarding the successful expansion of autologous $\alpha\beta$ T cell products from solid organ transplant recipients as follows:

“Both autologous and allogeneic virus-specific T cell therapies have been or are currently being tested in phase I trials in SOTR^{15,17,42} (CTRN12621000323820) or in a phase III trial in HSCTR (TRACE study: NCT04832607). Among these two strategies, the autologous approach is particularly attractive in SOTR to avoid the risk of allosensitization deleterious to the transplant⁴³. However, whether functional and CMV-reactive DOT cells can be amplified from immunosuppressed transplant recipients remains to be evaluated.”

8. l. 542: What was the rationale to exclude glucocorticoids from the study, although they are standard immunosuppressants used after SOT? Please add data on the effect of glucocorticoids on the cell product.

Reply: As requested, we have tested the impact of glucocorticoids (methylprednisolone) on DOT cell proliferation and anti-CMV reactivity in vitro. At the concentrations used in the maintenance treatment of SOTR (at around 10 ng/ml 12 hours after the ingestion) (PMID: 9797792), methylprednisolone did not inhibit the persistence and reactivity of

DOT cells against CMV-infected fibroblasts. These new data are now presented in **Supplementary Figure 8** of the revised manuscript.

9. Please add data on the killing capacity of patient derived T cell products regarding killing of the CMV+ fibroblasts.

Reply: Demonstrating the killing capacity of DOT cells against CMV-infected fibroblasts is technically challenging. Indeed, CMV infection is lytic and eventually leads to fibroblast death independently of T cell cytotoxicity. 5 days of infection, when DOT cells are cocultured with fibroblasts, is the limit to keep infected cells reasonably alive, making it technically unfeasible to distinguish additional killing mediated by DOT cells. Moreover, killing infected cells is not necessarily the most important function of $\gamma\delta$ T cells in the context of CMV infection. Secreting IFN γ , which has potent anti-viral functions and inhibits CMV replication is also critical to control the virus propagation.

Reviewer #3 (Remarks to the Author):

The paper by Marsères et al. asks whether the long-established capacity of human non-V δ 2+ T cells to respond to CMV-infected cells (and, at the same time, tumor cells) can be exploited as a cell therapy using the DOT cell technique to propagate and enrich for V δ 1+ T cells. Overall, the results seem encouraging and provide a basis for considering a clinical trial to limit CMV infection or reactivation in immunosuppressed transplant recipients. This is a major unmet need although analogous clinical approaches using $\alpha\beta$ T cells are further advanced. For the most part, the experiments are logical and seem well-executed, but there are areas where the authors' interpretations need to be placed very clearly in the context of DOT cells since they may not apply to human $\gamma\delta$ T cells in vivo. Indeed, this is an immunotherapy paper rather than a paper about human immunology, which is fine, so long as the two are not confused.

Main points:

1. Figure 1: this shows that DOT cells can be expanded equally well from CMV-sero(+) donor and CMV-sero(-) donors, thereby showing that an initial status of terminally differentiated effector non-V δ 2+ cells in CMV-sero(+) donors does not prevent the cells expanding and adopting a phenotype more like activated central memory cells. However, a major advantage of $\gamma\delta$ T cells for immunotherapy is their lack of MHC restriction and potential to be used in the allogeneic setting, in which case, a set of CMV-sero(-) donors would presumably be the main source. Hence, why is it important to know that cells from CMV-sero(+) donors can be expanded? The authors might argue that this is because autologous cells may be required for solid-organ recipients, but for that setting, which is a major unmet need, the authors need to be much more clear about the advantages over autologous $\alpha\beta$ T cells.

Reply: We thank the reviewer for this comment. As correctly pointed out, the relevance of expanding DOT cells from CMV-seropositive donors depends on the therapeutic

setting (allogeneic or autologous). At this stage, both allogeneic and autologous DOT cell therapies can be considered.

In the allogeneic setting, access to a broad and diverse donor pool is essential. CMV prevalence exceeds 50% in most western countries, reaching for instance over 70% in Spain and Portugal (PMID: 38275940). Demonstrating that DOT cells can be efficiently expanded from CMV-seropositive donors ensures broad donor availability, making it feasible to manufacture a diverse and partially HLA-matched donor bank. This could reduce the risk of allosensitization and more specifically the occurrence of donor-specific antibodies in transplant recipients.

In the autologous setting, refractory CMV infections typically occur in CMV-seronegative patients (transplanted with an organ coming from a CMV-seropositive donor) whose adaptive immune response to CMV is lacking (PMID: 26054538). In this context, generating DOT cells from the $\gamma\delta$ T cell pool of these patients may offer an advantage: naïve $\gamma\delta$ T cells can be expanded ex vivo into functional and CMV-reactive DOT cells, whereas it is challenging to obtain CMV-specific memory $\alpha\beta$ T cells from naïve cells (PMID: 25925682).

These points are now included in the Discussion section of the revised manuscript.

2. Figure 2: this shows that DOT cells from sero(+) or sero(-) donors limit CMV dissemination in vitro. However, the data do not seem as simple as the text describes. First, there is rather marginal activation of cytolysis by CMV-infected fibroblasts compared to medium alone, whereas the strong activation of cytolysis against infected macrophages is barely greater than that against mock-infected macrophages. For putative DOT-immunotherapy the fibroblast data might question efficacy, whereas the macrophage data might question safety. The IFN γ data seem more convincing.

Reply: We agree that IFN γ secretion assays provide a more convincing indication of DOT cell anti-CMV reactivity compared to CD107a assays. However, we would like to argue that the combined effects of IFN γ secretion and the limited, yet statistically significant, cytotoxic activity likely act synergistically to limit CMV dissemination, as illustrated in **Figure 2F**.

Regarding the safety concern raised about macrophage targeting, we would like to clarify that the macrophages used in these experiments are M1-like activated macrophages, generated by differentiating monocytes in the presence of GM-CSF for six days. These activated macrophages differ substantially from resting tissue macrophages, which mitigates potential safety issues raised by the reviewer. Importantly, this differentiation and activation step is necessary to render macrophages permissive to CMV infection and may explain why they induce a basal activation of DOT cells. This notwithstanding, infection by CMV significantly further increased DOT cell degranulation and IFN γ secretion, underscoring the capacity of CMV to activate DOT cells.

3. The specificity for CMV versus other viruses is striking, but this needs to be considered in the context of Fig 3.

Fig 3: this shows that the TCR is not required for targeting CMV-infected cells. Note, the abstract states: “DOT cells efficiently controlled CMV dissemination in vitro. This antiviral effect was independent of the $\gamma\delta$ TCR”. This seems very surprising given the wealth of evidence from this team that the TCRs of human non-V δ 2 T cells recognise self-antigens upregulated on CMV-infected cells. Furthermore, it was reasonable to assume that such TCR reactivities would account for CMV-specificity (above).

Reply: This issue of human non-V δ 2 T cells specificity for CMV is undoubtedly an interesting and striking point. This specificity is observed both in patients (only CMV infection can induce non-V γ 9V δ 2 T cell response and expansion, not HSV or VZV; Déchanet et al., JID 1999; Pitard et al., Blood 2008; Kaminski et al., JID 2021) and in vitro (only CMV infection can induce fibroblast ability to activate non-V γ 9V δ 2 T cells; Halary et al., JEM 2005; Kaminski et al., JID 2021). However, it is not clear at all whether TCR reactivities account for CMV-specificity. We have for instance shown in one of our study (Willcox et al., Nat Immunol 2012) that the V γ 4V δ 5 T cell clone LES was recognizing EPCR, a constitutively expressed ligand on fibroblasts. The specificity of this clone for CMV-infected cells was dependent on the strong induction of CD54/ICAM-1 expression by CMV. Hence, CMV specificity is most probably due to its capacity to induce a multimolecular stress signature on infected cells recognized not only by the TCR but also by co-stimulatory molecules such as LFA-1. Why HSV-1 and VZV cannot induce the same signature is currently not resolved, but might rely on inhibitory signals preventing $\gamma\delta$ T cell activation.

Regarding the TCR-independent CMV recognition by DOT cells, we are not as surprised as the reviewer. All our previous studies were conducted using non-V δ 2 T cell clones, or non-V δ 2 T cells isolated from CMV-infected transplant patients (which repertoire is highly restricted or even oligoclonal in some patients, Dechanet et al., JCI 1999). On the contrary, the repertoire of DOT cells has been shown to be highly polyclonal (Di Lorenzo et al., Cancer Immunol Res 2019, PMID: 30894378) supposedly because they expand from naïve cells. Then, it would have been more surprising to observe TCR-specificity of a polyclonal population. Tumor cell recognition by DOT cells was also previously shown to be TCR-independent by the team of B Silva-Santos (Di Lorenzo et al., Cancer Immunol Res 2019).

4. So, do the data hold up? The use of a blocking antibody is problematic because without a clear control to prove otherwise, it is common for anti-TCR antibodies, even those that can block, to cross-link TCRs, thereby clouding the interpretation. Therefore, the logical experiment is to delete the TCR, which is accomplished. Here the problem may be that (according to the Methods) the TCR deletion was validated over a 4-day period after which cells were rested overnight, and then used in the experiments. Prior to this, the cells had constant TCR stimulation for over two weeks and probably therefore carry a legacy of this exposure in terms of downstream signalling effectors and epigenetic imprinting.

Reply: We fully agree with the reviewer's interpretation. We had attempted to address this point in the discussion, but our explanation may not have been sufficiently clear. We did not mean that TCR signaling is not necessary for the overall DOT cell response. Rather, our data indicate that during the interaction with CMV-infected cells, TCR engagement is not necessary for DOT cell activation, as evidenced by the robust response of TCR-knockout DOT cells to CMV-infected, compared to mock-infected targets (**Fig. 3D** and **Fig. S5B** of the revised manuscript). In other words, the specific recognition of CMV-infected cells by DOT cells does not require ligand recognition by the TCR. We propose that the DOT cell expansion procedure, which involves constant TCR stimulation, provides an essential initial TCR signalling (that could be considered as a priming), and that CMV-infected cells then provide the costimulation, leading to a CMV-specific response.

To clarify this distinction in the manuscript, we have revised our wording by replacing "TCR-independent response" by "TCR-independent recognition" or by saying that TCR-engagement is not required during the interaction with CMV-infected cells.

5. This could explain why the evident reduction in IFN γ response (Fig 3E) is not complete. The hypothesis that TCR signalling is required could explain why there was no impact of blocking any NKR (Fig 3G) and seems to be supported by the data in Fig 4 (below).

Also, the argument that the high background of IFN γ in DOT cells responding to mock-transfected cells probably relates to a legacy of continued TCR signalling does not seem consistent with there being no background IFN γ in cells stimulated with isotype antibodies as a control for anti-CD3 (compare Fig 3D and 3E).

Reply: We respectfully disagree with this comment and would like to clarify the data as follows:

- In **Fig. S5B** of the revised manuscript (previously Fig. 3E), we can see that the background of IFN γ production by control DOT cells is comparable between "medium" alone and "mock" conditions, indicating that coculture with mock-infected fibroblasts does not induce a consistent additional background activation in these 3 DOT donors.
- In **Fig. S5A** of the revised manuscript (previously Fig. 3D), there is indeed background IFN γ secretion in the "Isotype" control condition, but because of the scale, it is not visible on the graph. When the same data are plotted on the same axis (as shown in **Figure R5** below), it becomes visible that IFN γ background levels are in the same range in the "isotype" condition of **Fig. S5A** and in the "medium" and "mock" conditions of **Fig. S5B**.

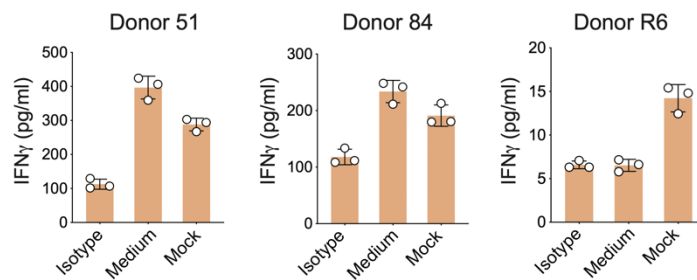


Figure R5: IFN γ secretion by DOT cells from 3 independent donors (51, 84 and R6), incubated for 24h in medium alone, with an isotype control antibody, or with mock-infected fibroblasts.

This basal secretion of IFN γ is consistent with the “legacy of continued TCR signaling” argued by the reviewer and that we fully agree with. The abrogation of this IFN γ production background in TCR-KO DOT cells allows to reveal the “true” CMV-specific response, as shown in the green histograms in **Fig. S5B** and **Fig. 3D** of the revised manuscript. Notably, the fold increase in IFN γ production by TCR-KO DOT cells in response to CMV-infected cells exceeded 15-fold (**Fig. 3D**), suggesting that the TCR-independent signals induced by CMV-infected cells are very potent and may play a crucial role in their activity.

6. Fig 4: this claims to show that LFA1 is important for the response to CMV-infected cells. This would be consistent with the TCR engaging CMV-infected targets and activating LFA1 via inside-out signalling, thereby enhancing intimate effector-target cell contacts. One can hypothesise that in TCRk.o. cells, residual signalling events following exposure to chronic TCR signalling are sufficient to activate LFA1. Indeed, if the TCR was not involved, and none of the NKR examined in Fig 3G is involved, it is very challenging to understand how activating LFA1 alone could induce a response from T cells, since it is at best a co-stimulator and primarily an adhesion molecule.

Reply: We show in Fig. 3F of the revised manuscript that a recombinant ICAM-1-Fc protein or an agonist anti-LFA-1 antibody can activate DOT cells even when used alone. This demonstrates that LFA-1 engagement, independently of TCR engagement, can activate DOT cells. LFA-1 engagement by CMV-infected cells might therefore be sufficient to activate TCR-KO DOT cells. It is possible that this response requires a prior activation of the $\gamma\delta$ TCR during DOT cell expansion, that would activate LFA-1 via inside-out signalling. We however cannot exclude the involvement of other undetermined activating receptor(s) during the co-culture with CMV-infected cells.

7. It is perhaps important to note that if the TCR were not required, the authors would be obliged to explain why transfer of $\gamma\delta$ T cells would be preferable to transferring NK cells. In fact, the response of donor 2 LFA1k.o. cells (Fig 4E) is interpreted as negative, but it

looks equally as strong as the responses of donors 1 and 2 TCRk.o. cells (Fig 3E) which were scored as positive.

Reply: Regarding the adoptive transfer of $\gamma\delta$ T cells versus NK cells, please see our response to comment 9 below.

It is difficult to compare the response of DOT cells between Figure 3 and Figure 4 of the original manuscript, because they have been generated from different donors. As the reviewer can appreciate in **Fig. 2B**, IFN γ production is very variable from one DOT cell product to another.

We would like to mention here our reply to **Reviewer 1 comment 29**:

Reply: We appreciate this important comment and fully agree that conclusions drawn from a single donor with efficient knockout are insufficient. To address this comment, we attempted LFA-1 knockout in DOT cells from three additional donors. Knockout efficiency was assessed by flow cytometry (**Figure R4**).

Unfortunately, these three attempts to obtain a full knockout failed (as shown in **Figure R4 below**), as LFA-1 expression remained high following editing. We also attempted to enrich LFA1-negative cells by cell sorting, but in all cases, we were unable to recover a sufficient number of viable cells for downstream functional assays. Given these technical limitations, we have decided to remove the initial LFA-1 knockout data from the manuscript. Instead, conclusions about the involvement of LFA-1 in CMV-specific reactivity are now based on functional experiments using anti ICAM1 and LFA-1 blocking antibodies, now performed in six independent DOT donors in total. These results consistently support a role for LFA-1 in DOT cell anti-CMV reactivity.

We have revised **Figure 3** and **Figure 4** accordingly. The two figures are now merged in one. In the revised manuscript, we have mentioned this technical limitation in the Results section. We have also revised our interpretation and discussion of this mechanism accordingly (in the Discussion section).

8. In sum, it seems highly likely to this reviewer that human V δ 1 cells require the TCR to engage CMV-infected cells and that they then activate LFA1. In DOT cells, LFA1 is already activated by chronic TCR signalling and can collaborate with residual TCR signals in TCR k.o. cells. This is an unusual situation that should not be confused with human immunology.

Reply: We thank the reviewer for this summary that fits perfectly with our own interpretation. We agree that DOT cell activity cannot be compared to the physiological activity of non-V δ 2 T cells. As pointed out by the reviewer, our previous studies have consistently shown that non-V δ 2 T cells, including V δ 1 T cells, recognize CMV-infected cells in a TCR-dependent manner (Halary et al., JEM 2005; Willcox et al., Nat Immunol 2012; Kaminski et al., JID 2021).

DOT cells represent a highly activated cell product, shaped by in vitro expansion and chronic TCR engagement, leading to a unique way of responding to CMV-infected cells.

While this does not directly reflect the physiology of human non-V δ 2 T cells, it is essential to dissect the functional requirements of DOT cells to develop them as a therapeutic product, which was the purpose of our study. We have clarified this point in the revised manuscript (Results and Discussion sections).

9. Fig 5: this shows that murine $\gamma\delta$ T cells produced via a DOT-type culture can be transferred to recipient lymphocyte, ILC, and NK cell deficient mice, wherein they become activated and distribute into multiple organs to limit CMV in vivo. This is a proof-of-principle; one suspects that NK cells might have achieved the same outcome.

Reply: We agree with the reviewer that our in vivo experiments using murine $\gamma\delta$ T cells generated via a DOT-like protocol serve as a proof-of-principle.

NK cells indeed have well-established antiviral functions, including against CMV, even though the demonstrations have been mainly obtained in mice. However, there are at least two reasons why DOT cells may offer distinct advantages or complement NK-based therapies:

- We have shown that the DOT protocol allows for a robust and efficient expansion of clinically relevant numbers of non-V δ 2 T cells from the peripheral blood of both healthy individuals and immunosuppressed transplant recipients. In contrast, generating large-scale, functional NK cell products can be more variable and often requires longer culture times or feeder-cell systems, which pose regulatory and translational challenges (PMID: 38443448; PMID: 40681797).
- DOT cell harbour a unique landscape of receptors, notably lacking expression of NKG2A and KIRs (**Fig. 1G** and **Figure R6** below). These receptors are classically expressed by NK cells, contribute to their recognition of CMV-infected cells, and are common targets of CMV-induced immune evasion strategies (PMID: 12032324, PMID: 18069056, PMID: 29632329). Many evasion mechanisms have been selected by CMV to escape from NK cell control, which has not been reported yet to our knowledge for $\gamma\delta$ T cells or DOT cells. In addition, as discussed above, TCR priming during the expansion phase most likely participates in the overall response of DOT cells to CMV, distinguishing them from a NK cell product. Altogether, the molecular mechanisms underlying CMV recognition by DOT and NK cells are likely distinct.

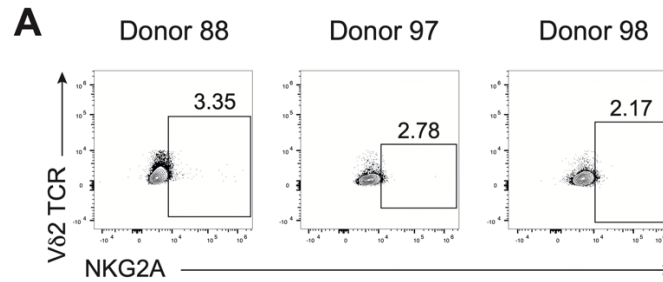


Figure R6: NKG2A expression on DOT cells.

DOT cells amplified from three independent donors (88, 97 and 98) were stained with antibodies directed against the populations markers CD3, $\gamma\delta$ TCR, V δ 1 TCR and V δ 2 TCR and the NK receptor NKG2A. Shown are percentages of NKG2A⁺ V δ 2^{neg} $\gamma\delta$ T cells. Fluorescence Minus One controls were used to set positivity gates.

Fig 6: this shows that the DOT protocol can be applied to kidney transplant recipients and that the cells may be functional in the presence of common immunosuppressants that would be the milieu for an adoptive therapy. This is encouraging although no evidence is given for how the cells remain functional in the immunosuppressants.

Reply: Cyclosporin and Tacrolimus are calcineurin inhibitors that block NFAT translocation and IL-2 production in T cells. However, the DOT protocol does not rely on IL-2. Instead, it relies on IL-15 as the critical cytokine during the second stage of expansion and differentiation (day 7 to day 21 of expansion). Moreover, IL-15 has been shown to enhance DOT cell therapeutic efficacy in vivo (Sanchez Martinez et al., J Immunother Cancer 2022, PMID: 36162920; Blanco-Dominguez et al., Nat Cancer 2025, PMID: 40240620). Importantly, IL-15 signalling is independent of the calcineurin-NFAT pathway. Based on these observations, we hypothesize that DOT cells are more dependent on IL-15 than IL-2 signaling and are therefore resistant to the effects of calcineurin inhibitors.

Regarding Everolimus, an mTOR inhibitor, we have previously shown that, at clinically relevant concentrations such as used in a transplantation setting, everolimus does not prevent T cell proliferation nor response to CMV, and by contrast promotes T cell survival and prevents T cell dysfunctionality (Kaminski et al., JASN 2022, PMID: 34725108). Consistently, Everolimus has been associated with a reduced occurrence of CMV infections in transplant recipients in several independent studies (PMID: 34990047; PMID: 31152476; PMID: 25988935).

We have summarized these points in the discussion section of the revised manuscript.

Reviewer #4 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review

and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Point-by-point response

Reviewer #1 (Remarks to the Author):

Reviewer Response to Rebuttal Letter (Reviewer #1 + co-reviewer) :

We would like to begin by congratulating the authors on the work they have put into revising their manuscript. The additional experiments, clarifications to the figures, and improvements to the text have significantly enhanced the quality of the work. We especially appreciate the consistency in which CMV status of the donor cells is now shown in the figures, providing transparency in the functional equivalence of the DOT cells expanded from different donors. These thoughtful revisions demonstrate a strong commitment to scientific rigor and communication, and we commend the authors for their efforts.

[Reply: We thank the reviewers for their kind comments.](#)

That said, we would like to kindly request the authors to address a few remaining points:

1. Inclusion of Reviewer-Specific Figures

Several figures created specifically in response to reviewer comments (the CRISPOR output data with sgRNA specificity scores, and specifically R1 and R3) provide valuable insights and clarity. We believe these should be included in the supplemental data to ensure transparency and to benefit future readers. Figure R1 addresses baseline gating data often requested in the field for assay validation purposes. Furthermore, figure R3 shows important data serving as a positive control for main figure 3B, without which the main figure is not adequately interpretable.

[Reply: All requested figures have now been included in the supplementary materials of the revised manuscript. The gRNA characterization table \(previously Figure R3\) was included as **Supplementary Table 2**. Figure R1 \(Fluorescence Minus One controls from the different culture conditions\) was included as **Supplementary Figure 4b**. Finally, Figure R3 \(positive control experiment for V \$\delta\$ 1 TCR blocking\) was included as **Supplementary Figure 5a**.](#)

2. Clarification on Specific Findings

We would appreciate further commentary on the following points:

- Comment #14 and #39: While the authors have touched on the presence of IL-18 (in CMV-infected human endothelial cells) and IL-15 (in bone marrow of CMV-infected individuals), a more detailed contextualization would strengthen the author's views on

what this means regarding the survival and activity of DOT cells in human patients. For instance, which concentration of these cytokines is required for the activity/support of these cells *in vitro*, and how does that correlate with expected levels in relevant tissues in a CMV-infected SOTR patient receiving the DOT therapy?

Reply: In our hands, the optimal concentration of IL-15 required to support DOT cell proliferation *in vitro* is 70 ng/mL, while the concentration of IL-18 needed *in vitro* to potentiate anti-CMV activity is 50 ng/mL. To the best of our knowledge, cytokine levels (including IL-18 and IL-15) have never been measured directly in CMV-infected human tissues. However, serum concentrations of IL-18 increase during CMV infection, reaching median levels of 2000 pg/mL (PMID: 34216086, reference previously included in the Discussion, line 375). Taken together, while systemic IL-18 levels in patients are expected to be lower than the concentrations used in our *in vitro* assays, local microenvironments (tissue niches, infected organs) may achieve higher, functionally relevant cytokine concentrations.

- Figure R3 and main figure 3B: We do not fully agree with the author's interpretation of these results. Although from later experiments using functionally knocked-out $\gamma\delta$ TCR cells, it can be concluded that DOT cells can recognize CMV-infection in the absence of $\gamma\delta$ TCR, the data presented in R3 and 3B are not contributing to this finding. Specifically, in R3, the authors show the inhibitory effect of the V δ 1 antibody, which is reducing IFN γ induction by approximately 50% (and sometimes less). We appreciate the significant reduction compared to iso control, but the blocking effect seems incomplete, and residual δ 1 signaling effects cannot be excluded. Combined with the data presented in figure 3B, where we see variability of DOT cell response to CMV infection in combination with V δ 1 blocking antibody on the individual donor level, with most donors showing a decrease in IFN γ secretion after V δ 1 blocking, and 2 donors showing an opposite effect. Grouped together, these changes are not deemed statistically significant different ($P=0,051$), however in the context of incomplete V δ 1 blocking and the large variability observed, the validity of this conclusion can be debated (i.e. the lack of difference may be because residual δ 1 signaling is sufficient to maintain a certain level of IFN γ secretion)

Reply: We agree that in Figure R3 (now **Supplementary Figure 5A** of the revised manuscript) the blocking effect is incomplete. As the reviewers point out, residual V δ 1 TCR signaling could result from this incomplete blocking. We have therefore modified our interpretation of these results and conclusions as follows (in the Results section, line 182): «In contrast to these observations, adding a V δ 1 TCR blocking antibody to cocultures did not impact IFN γ secretion by DOT cells in response to CMV, as shown for eight independent donors in **Fig. 3b**. However, as this result could have been explained by an incomplete V δ 1 TCR blocking (see **Figure. S5a**), we undertook to knock out the $\gamma\delta$ TCR of DOT cells obtained from three independent donors”.

- Figure 3H: This result raises an interesting observation that merits a brief discussion, even if speculative – now that the CMV status of the donor cells from which the DOTs are

expanded is made visible, it seems that a trend exists of higher IFN γ induction in CMV + IL-18 cocultured DOTs from CMV negative donors (ie: gray dots are predominantly in the upper values, while the red dots dominate the lower values). Do the authors wish to comment on this interesting trend?

[Reply:](#) A trend is indeed visible in **Fig. 3h**. This trend is visible in the absence of any blocking, and remains consistent throughout blocking conditions. However in **Fig. 2c**, the impact of the DOT cell donor CMV serotype on CMV-induced IFN γ secretion was specifically addressed in a larger cohort of donors (n=13 CMVneg and n=14 CMVpos). This comparison showed no significant difference in between CMVneg and CMVpos donors.

- Figure 3G: We appreciate the added clarity of activating and blocking antibodies to entire figure 3. Specifically in figure 3G, the activating label underneath ICAM-1 is missing. Please add the missing label for additional clarity.

[Reply:](#) We have clarified **Fig. 3g** to indicate that DOT cells were stimulated with recombinant human ICAM-1, indeed activating LFA-1 on DOT cells. This was also clarified in the Results section (lines 215-216).

- Lastly, we encountered these 2 minor points:

1. the HSCTR acronym is explained in line 49, but used earlier without an explanation in line 31 and 33.

[Reply:](#) We have now defined the acronym HSCTR when first used in the revised manuscript (line 31).

2. We noticed that in the abstract and other places in the main text, 'amplified' is changed to 'expanded', which fits better in the context of proliferating cells. The authors should consider making this same change to line 65 and 117, where 'amplified' is still used.

[Reply:](#) We have now systematically replaced 'amplified' by 'expanded' when describing DOT cells in the revised manuscript.

We consider these points to be addressable, and we look forward to seeing them incorporated in the final version.

Once again, we thank the authors for their work and responsiveness throughout the review process.